17 At the start of year n, the quantity of a radioactive metal is P_n
At the start of the following year, the quantity of the same metal is given by

$$P_{n+1} = 0.87P_n$$

At the start of 2016 there were 30 grams of the metal.

What will be the quantity of the metal at the start of 2019?
Give your answer to the nearest gram.

2017:30 × 0.87 = 26.1g.

1 mark
First year

2018:26.1 × 0.87 = 22.707g.

2019:22.707 × 0.87 = 19.75509g.

1 mark
Three yearly steps

20g (nearest gram).

1 mark
20g rounded

.....grams

(Total for Question 17 is 3 marks)